HOW X, Y & Z GRADIENTS CREATE AN IMAGE

GE MRI

From Slice Selection to Frequency, Phase and Spatial Encoding

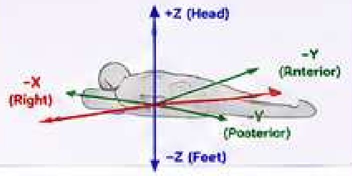
Gradients work together in time to localize the MR signal in 3D space. Each axis has a specific job.

X = Frequency / Readout (Left ↔ Right) Y = Phase Encoding (Anterior ↔ Posterior) Z = Slice Selection (Head ↔ Foot)

KEY TAKEAWAYS

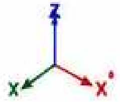
- Z selects a slice.
- Y steps through lines (phase encode).
- X samples each line (frequency encode).
- RF excites. Gradients encode.
- All three axes together build k-space.
- k-space is transformed to form the image.

AXIS CONVENTION (Patient Supine)

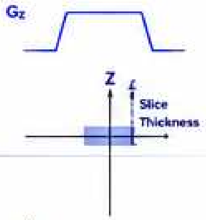


1. THE THREE GRADIENTS – WHAT THEY DO

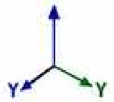
Z GRADIENT (Slice Selection)



- Applied during RF pulse
- Varies linearly along Z
- Makes Larmor frequency different with position
- Only spins in the selected slice are excited

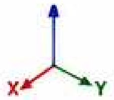


Y GRADIENT (Phase Encoding)

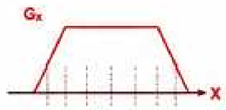


- Applied for a short time
- Imparts a position-dependent phase along Y
- Steps up between TRs
- Selects one line of k-space each TR

X GRADIENT (Frequency / Readout)

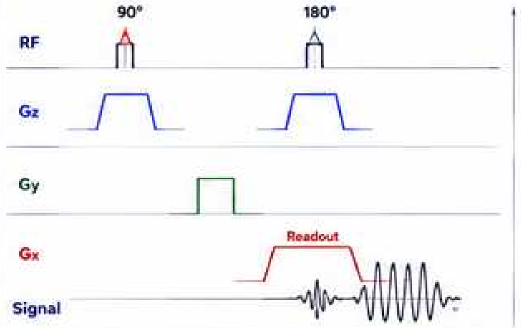


- Applied during readout
- Varies along X while we acquire the signal
- Determines frequency of the signal across X



2. HOW THE THREE GRADIENTS WORK TOGETHER (TIME LINE)

Typical 2D Spin Echo Sequence (One Slice)



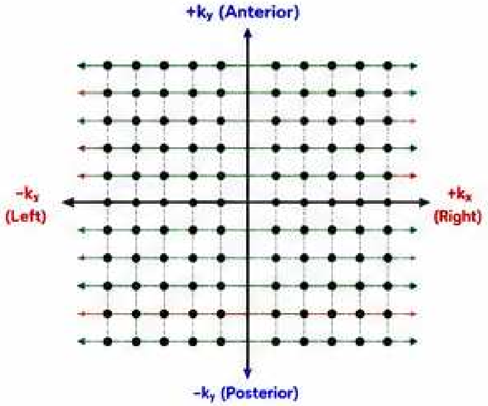
1. 90° RF + Gz (slice select): Excites a thin slice.
2. Gy (phase encode): Adds a phase that depends on position along Y.
3. Gx (frequency encode): Readout gradient. Signal is acquired across X.
4. Repeat with next Gy level: Moves to next line of k-space.
5. After all phase steps: Entire k-space is filled. Image is reconstructed.

What Happens

1. Slice is selected along Z.
2. Phase differences are created along Y.
3. Frequencies are mapped along X while we sample.
4. Y is stepped from $-k_y$ (top) to $+k_y$ (bottom).
5. k-space is complete → Image is formed.

3. BUILDING k-SPACE (2D)

k-space is filled line by line.



Each green line is acquired in one TR with a different Gy (phase encode) level.

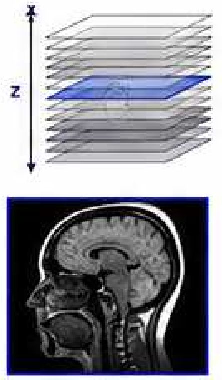
- Horizontal (X) = Frequency (Readout)
- Vertical (Y) = Phase Encode
- Center of k-space (low spatial frequency) contains image contrast.
- Outer k-space (high spatial frequency) contains detail.

More phase lines = Higher resolution in Y direction

4. VISUALIZING THE ROLES OF X, Y & Z

Z: Slice Selection

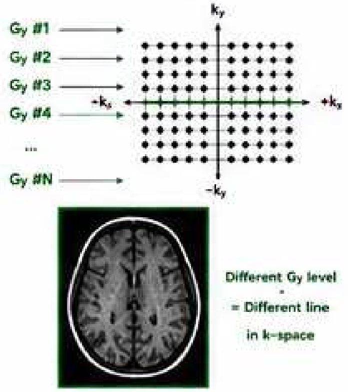
Only the spins in the selected slice are excited.



Different Z position = Different slice

Y: Phase Encoding

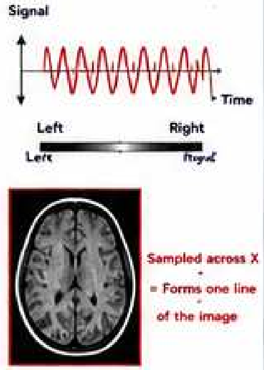
Each Gy step selects a different line in k-space.



Different Gy level = Different line in k-space

X: Frequency Encoding

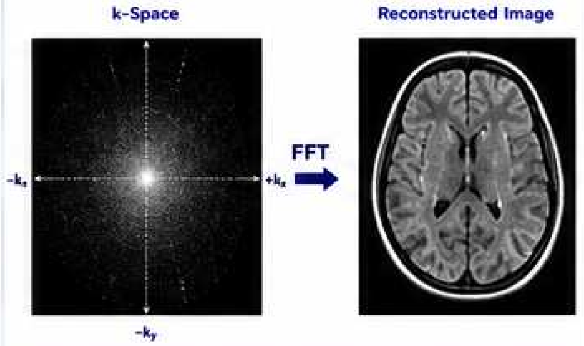
Gx reads out the signal across the slice.



Sampled across X = Forms one line of the image

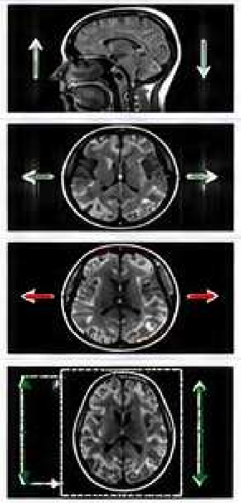
5. FROM k-SPACE TO IMAGE

After all phase lines are acquired:



The image you see is the result of a Fourier Transform (FFT) of the complete k-space data.

6. WHAT HAPPENS WHEN SOMETHING GOES WRONG?



Problem	Likely Cause	Image Appearance	What to Check
Wrong slice / Slice shift	Gz offset / miscalibration		• Gz offset • Shim • Gradient calibration
N/2 Ghosting (EPI)	Gy polarity mismatch, timing error, eddy currents		• Gy polarity • Pre-emphasis • Eddy current comp.
Geometric Distortion (Left-Right)	Gx nonlinearity, delay, eddy currents, wrong polarity		• Gx calibration • Pre-emphasis • Gradient delay
FOV / Scaling Error	Gradient amplitude calibration error		• Gradient gain • FOV settings • Calibration

ENGINEER NOTES

- Always change one thing at a time.
- Use small FOV and adjust offsets.
- Verify gradient polarity with test tools.
- Check pre-emphasis and gradient delay.

ABBREVIATIONS

RF = Radio Frequency	TR = Repetition Time	k-space = Spatial Frequency Space
GPA = Gradient Power Amplifier	TE = Echo Time	Gx = Gradient X (Left-Right)
FOV = Field of View	FFT = Fast Fourier Transform	Gy = Gradient Y (A-P)
		Gz = Gradient Z (Head-Foot)

SAFETY REMINDER

- Gradients produce strong forces and heat.
- Do not disable safety interlocks.
- Follow GE safety procedures.

THE BOTTOM LINE

RF excites. Gradients encode.

X reads (frequency). Y steps (phase). Z selects (slice).

All three together build k-space → Image.

GE HealthCare
Imagination at work